Fabian Moritz Fuchs

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Academic Positions

Postdoctoral Research Fellow, Luiss Guido Carli, Department of AI, Data and Decision Sciences, Sep 2025 - now.
Research Associate, Bielefeld University, Center for Mathematical Economics (IMW), Dec 2022 - Aug 2025.
Researcher – Global Skills Strategy Program, University of Waterloo, Department of Statistics & Actuarial Science, Jun 2025.
Teaching Assistant, University of Konstanz, Department of Mathematics, Apr 2022 - Aug 2022.
Research Assistant, University of Konstanz, Department of Economics, Apr 2021 - Mar 2022.
Research Assistant, University of Konstanz, Department of Mathematics, Apr 2021 - Oct 2021.
Research Assistant, San Francisco State University, Lam Family College of Business, Mar 2020 - May 2020.

Education

Doctor of Economics (Dr. rer. pol.), Bielefeld University, 17.04.2026.
Master of Science (M.Sc.), Financial Mathematics, University of Konstanz, 10.10.2022.
Bachelor of Science (B.Sc.), Financial Mathematics, University of Konstanz, 22.04.2021.

Publications and Preprints

[1] F. Fuchs, *Viscosity Solutions for Hamilton-Jacobi equations on Separable Banach Spaces*, 2026. arXiv: 2610.18496 [math.AP].

[2] D. Criens and F. Fuchs, *Risk-sensitive exit-time control for stochastic differential equations with path-dependent coefficients*, 2026. arXiv: 2607.18192 [math.OC].

[3] F. Fuchs and M. Nendel, "Existence of viscosity solutions to abstract Cauchy problems via nonlinear semigroups," *Bulletin of the London Mathematical Society*, vol. 58, no. 5, e70386, 2026. DOI: https://doi.org/10.1112/blms.70386

[4] G. Bolli and F. Fuchs, *Projected Evolutionary Lifting and Well-Posedness of Stationary Hamilton-Jacobi-Bellman Equations in Infinite Dimensions*, 2026. arXiv: 2605.05028 [math.OC].

[5] F. Fuchs, "New Perspectives on Existence and Uniqueness of Viscosity Solutions," Thesis, Fakultät für Wirtschaftswissenschaften, Institut für Mathematische Wirtschaftsforschung, 2026, p. 157. DOI: https://doi.org/10.4119/unibi/3016326

[6] S. Della Corte, F. Fuchs, R. C. Kraaij, and M. Nendel, "A comparison principle based on couplings of partial integro-differential operators," *Journal de Mathématiques Pures et Appliquées*, vol. 210, no. 103872, pp. 1–54, 2026. DOI: https://doi.org/10.1016/j.matpur.2026.103872

[7] S. D. Corte, F. Fuchs, R. C. Kraaij, and M. Nendel, *A Strict Comparison Principle for Integro-Differential Hamilton-Jacobi-Bellman Equations on Domains with Boundary*, 2025. arXiv: 2512.04005 [math.AP].

Selected Talks

Mathematical AI Seminar (The Fields Institute), 10.07.2026. *Invited Talk.*
XIII Bachelier World Congress of the Bachelier Finance Society (Università degli studi di Bologna), 02.07.2026.
Oberseminar Stochastik (Universität Freiburg), 17.06.2026. *Invited Talk.*
Mathematical Models for the Impact of Human Activities on the Environment and Economic Decision Making (Luiss Guido Carli, Rome), 02.10.2025.
ISIPTA 2025 (Bielefeld University), 15.07.2025.
12th General AMaMeF Conference (University of Verona), 25.06.2025.
Mathematics of Uncertain Systems for Economics and Finance (Rimini), 26.05.2025.
Risk and Uncertainty in Economics and Finance (University of Johannesburg), 04.04.2025.
17th German Probability and Statistics Days (TU Dresden), 12.03.2025.
FDM-Seminar (Universität Freiburg), 25.10.2024. *Invited Talk.*
BGTS Doctoral Day 2024 (Bielefeld University), 11.10.2024.
7th Berlin Workshop on Mathematical Finance for Young Researchers (HU Berlin, TU Berlin), 05.09.2024.
12th Bachelier World Congress of the Bachelier Finance Society (FGV EMAp, Rio de Janeiro), 10.07.2024.

QED Jamboree 2024 (Universiteit van Amsterdam), 25.05.2024.
QFW 2024 (Università degli studi di Bologna), 11.04.2024.

Teaching

Quantitative Methods for Management, Luiss Guido Carli, Teaching Assistant, Winter Term 2026/2027.
Finance 2, Bielefeld University, Teaching Assistant, Summer Term 2024.

Tutorials in *Probability Theory; Stochastic Processes; Corporate Finance.*

Awards, Grants, and Third Party Funding

BFS Travel Award (300€) of the Bachelier Finance Society, April 2026.

Research grant *Impact of the Human Activities on the Environment and Economic/Financial Decision Making: Mathematical Models through Stochastic* (ssd STAT-04/A), Scientific Director Prof. Fausto Gozzi, Sep 2025 - Mar 2028.

Best Course Award of the Faculty for Business Administration and Economics, Bielefeld University, for *Tutorial Finance 2*, Jul 2024.

Investigator in Project C7 *Markovian dynamics under model uncertainty* of Bielefeld University's CRC 1283, funded by the German Research Foundation (Deutsche Forschungsgemeinschaft), Dec 2022 - Aug 2025.

Organization and Support of Academic Events

ISIPTA 2025 (ZiF, Bielefeld), member of the organizing committee, 15.07.2025 - 18.07.2025.
Mathematics of Uncertain Systems for Economics and Finance (Hotel Sporting, Rimini), member of the organizing committee, 26.05.2025 - 30.05.2025.
19th BiGSEM Workshop (ZiF, Bielefeld), organizer, 16.12.2024 - 17.12.2024.
11th General AMeMaF Conference (Ravensberger Spinnerei, Bielefeld), member of the local organizing committee, 26.06.2023 - 30.06.2023.

Other Academic Positions

Member of the Board of the Center for Mathematical Economics (IMW), Bielefeld University, Apr 2023 - Aug 2025.

Referee Activities

Mathematics and Financial Economics; SIAM Journal on Financial Mathematics; Journal of Optimization Theory and Applications; International Journal of Approximate Reasoning; Bollettino dell'Unione Matematica Italiana; Nonlinear Differential Equations and Applications (NoDEA)

Supervised Master's Theses

Nina Brinkhaus (supervisory support for Max Nendel, 2024); Kristina Wegner (supervisory support for Max Nendel and Frank Riedel, 2024).

Other Positions

IT Support of the Center for Mathematical Economics (IMW), Bielefeld University, Nov 2023 - Aug 2025.

Languages

English (C2), German (C2), Italian (A1)

Date: October 14, 2026